

From Narrative to Alpha

Exogenous Perception Pressure as an Orthogonal Signal in an
Autonomous Microstructure Alpha Book: Timescale Separation,
Effective Breadth, and the Adverse-Selection Barrier

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Abstract

A book of alpha signals built entirely from order-book microstructure is powerful but structurally narrow: its signals share one information source (order flow), occupy one band of timescales (sub-second to minutes), and are jointly capped by an adverse-selection barrier that collapses a large unconditional information coefficient once one conditions on the states in which a passive fill actually occurs. We ask what a *single exogenous signal from a different information domain* contributes to such a book, and we answer at the level of methodology rather than measured performance. Taking *Perception Pressure* $P(K, t)$ —a signed, credibility-weighted media-narrative signal produced by the Prism system—as the exogenous input, we specify (i) a source-agnostic integration architecture in which a narrative series is aligned to market bars by a point-in-time as-of join and scored by the same information-coefficient, mutual-information, and five-gate replication machinery as any endogenous feature, and (ii) three theoretical reasons the addition can improve the book that do not require the exogenous signal to be individually strong. First, *timescale separation*: narrative momentum lives in a spectral band essentially disjoint from the 0.005–0.1 Hz band that carries order-book-imbalance predictability, so the two are close to orthogonal by construction. Second, *effective breadth*: under the fundamental law of active management the information ratio scales with the effective number of independent bets $N_{\text{eff}} = N/[1+(N-1)\bar{\rho}]$, and appending a low-correlation signal raises N_{eff} more than appending another correlated microstructure signal of equal strength. Third, *partial evasion of the adverse-selection barrier*: because a slow exogenous signal is harvested by patient execution over its own horizon rather than by reactive spread-crossing, its coupling to the fillable-state indicator—the mechanism that destroys microstructure edge—is weaker, so its conditional information coefficient need not collapse in the same way. Each claim is stated as a hypothesis with an explicit, leakage-controlled, false-discovery-rate-corrected test; we report no performance figures. The contribution is a theory and a protocol for making an autonomous microstructure research platform *more complete*—a decorrelated, multi-domain alpha book that reports its own honest effective breadth—by admitting narrative as a first-class, gated signal source.

Keywords: exogenous signal, alpha combination, signal orthogonality, effective breadth, fundamental law of active management, adverse selection, mutual information, information coefficient, narrative, market microstructure

JEL Classification: C55, C58, G14, G17

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1 Introduction

1.1 The completeness problem of a single-domain alpha book

An autonomous microstructure research platform can, in principle, manufacture an unbounded number of features from the order book: imbalances, depth slopes, queue dynamics, trade-sign runs, spectral and information-theoretic transforms. In practice these features are far from independent. They are computed from the same underlying object—the limit order book and the trade tape—so they share information, co-move, and, when combined, deliver far less than the naive sum of their individual information coefficients. Three structural facts bound such a book:

1. **One information source.** Every signal is a functional of order flow. Whatever order flow cannot see—an exogenous shock forming in public discourse before it reaches the tape—no such signal can anticipate.
2. **One band of timescales.** Order-book-imbalance predictability is empirically concentrated at high frequency and decays to noise within minutes; a companion study on the same venue localizes essentially all of its predictive power to the 0.005–0.1 Hz band (periods of 10–200 s) with an Ornstein–Uhlenbeck mean-reversion half-life of a few seconds. A book living only in this band is exposed to a single regime of decay.
3. **A shared economic ceiling.** On a decentralized perpetual-futures venue, a strong *unconditional* predictor can be economically inert: conditioning the information coefficient on the states in which a directionally-correct passive fill would actually occur collapses it

from large to near zero. This adverse-selection barrier is a property of order flow itself and therefore afflicts order-flow signals *jointly*.

A book that is deep in one domain but empty in all others is thus fragile in a precise sense: its signals fail together, decay together, and are filled against together. Robustness—and additional, genuinely new information ratio—requires a signal that is *not* a function of order flow.

1.2 Narrative as the exogenous domain

The natural exogenous domain for a liquid, sentiment-driven asset class is public narrative. We take as given the *Perception Pressure* signal $P(K, t)$ and its momentum $\dot{P}(K, t)$ defined in the companion instrument paper (Onat, 2026a): a per-concept, signed, credibility-weighted media-narrative series, sampled on a fixed cadence, in which attention and directional sentiment are multiplicatively coupled and no single source can manufacture the measure. Our concern here is not how P is computed—that is the instrument paper’s subject—but what it is *worth to a trading book* and how to admit it without contaminating an otherwise disciplined research process.

1.3 Contributions

1. **A source-agnostic integration architecture** (Section 4) in which an exogenous narrative series is aligned to market bars by a strictly backward-looking as-of join and then scored—by information coefficient across horizons, by mutual information, and through a five-gate replication protocol—by exactly the machinery that scores an endogenous feature. The narrative signal is not privileged; it earns its place by the same gates.
2. **A timescale-separation argument** (Section 5) that narrative momentum and order-book-imbalance predictability occupy near-disjoint spectral bands, making the exogenous signal approximately orthogonal to the existing book by construction rather than by luck.
3. **An effective-breadth argument** (Section 6) that formalizes, via the fundamental law of active management and the effective number of independent bets, why a low-correlation signal contributes more marginal information ratio than another correlated microstructure signal of equal individual strength—and why the binding quantity is *stress-regime* correlation.
4. **An adverse-selection argument** (Section 7) that a slow exogenous signal, harvested by patient execution over its own horizon, has weaker coupling to the fillable-state indicator than a fast order-flow signal, and therefore need not suffer the same conditional-IC collapse.
5. **A falsifiable evaluation protocol** (Section 8) stating, for each claim, a leakage-controlled and multiple-testing-corrected test with explicit kill criteria. We report no empirical figures; this is a methods paper.

We are explicit throughout that the value of an exogenous signal is an empirical question. What this paper supplies is the theory that says *where* to look for that value, and the protocol that keeps the search honest.

2 Background and Related Work

Signal combination and breadth. Grinold’s fundamental law of active management (Grinold, 1989; Grinold and Kahn, 2000) relates the information ratio of a strategy to the skill of its signals and the number of independent bets, $IR \approx IC \cdot \sqrt{BR}$. When signals are correlated the naive breadth $BR = N$ overstates the achievable ratio; the operative quantity is

an *effective* breadth that discounts for redundancy. We use the effective-number-of-bets form $N_{\text{eff}} = N/[1 + (N - 1)\bar{\rho}]$ developed for this platform’s process framework (Onat, 2026c), which makes the marginal value of a new signal an explicit function of its correlation to the incumbent book.

Adverse selection and the limits of microstructure alpha. That a statistically strong microstructure predictor can be economically inert is documented for this venue in the companion microstructure-alpha study (Onat, 2026b), where the imbalance information coefficient of ≈ 0.45 at second horizons collapses to ≈ 0.03 once restricted to fillable states—an empirical instance of the classical adverse-selection problem of informed versus uninformed order flow (Glosten and Milgrom, 1985; Kyle, 1985). The barrier motivates the search for signals whose exploitation does not require winning a race for the same passive fills.

News, attention, and returns. That media tone and attention carry return-relevant information is long established (Tetlock, 2007; Da, Engelberg, and Gao, 2011); the instrument paper (Onat, 2026a) situates Perception Pressure within this literature. The present paper is downstream of all of it: we treat the existence of *some* narrative-return relationship as a hypothesis to be tested under controls, and study what its inclusion does to a microstructure book’s *portfolio* of signals.

Information-theoretic evaluation. We adopt mutual information as the currency of predictive content, justified through Fano’s inequality as a bound on achievable prediction error (Cover and Thomas, 2006), estimated by the k -nearest-neighbour method (Kraskov, Stögbauer, and Grassberger, 2004), and controlled for multiple testing by the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995). These are imported from the consuming platform, not invented here.

3 The Exogenous Signal

We recall only what is needed downstream. For a tracked concept K (an entity mapped to a tradable symbol, e.g. “Bitcoin” $\rightarrow$ BTC), the instrument produces on a fixed cadence a salience $A(K, t) \geq 0$, a valence $V(K, t) \in [-1, 1]$, a signed perception $P(K, t) = A(K, t) V(K, t)$, and its momentum $\dot{P}(K, t) = P(K, t) - P(K, t - 1)$.

Assumption 3.1 (Entity map). A fixed, time-invariant, publicly reproducible map ϕ sends tradable symbols to sets of tracked concepts, $\phi(\text{BTC}) = \{K_1, \dots\}$. When several concepts map to one symbol their signals are aggregated by a salience-weighted mean, itself computed only from information available at t .

Remark 3.1 (Momentum is the leading candidate). As argued in the instrument paper, a level P that merely co-moves with price is contemporaneous and of limited predictive use, whereas a narrative that is *building*, $\dot{P} > 0$, before it is fully priced is the economically interesting case. We therefore treat $\dot{P}$ (and higher-order changes) as the primary candidate and P , A , V as secondary. Nothing in the architecture depends on this choice; it is a prior about where predictive content is most likely, to be confirmed or rejected by the gates.

4 Integration Architecture

The design principle is that the exogenous signal must be *indistinguishable*, to the scoring machinery, from an endogenous feature—so that it is held to the same standard and enjoys no special pleading.

4.1 The feature-source contract

A narrative series enters through a minimal contract: a source exposes `load(symbol, start, end)` returning a data frame indexed by a UTC timestamp with one column per narrative component. The platform’s process runner merges these columns onto market bars and, from that point on, a narrative column is scored by `ic_horizon` and `mi_ksg` exactly as an order-book column is. Processes operate on columns and never learn a column’s origin. This is the formal sense in which narrative is a first-class signal: it is admitted by the type system, not by a bespoke code path.

4.2 Point-in-time alignment

Definition 4.1 (Backward as-of join). Let market bars close at times $\{t_i\}$ and let narrative snapshots be stamped at $\{\tau_j\}$, each τ_j being the instant the snapshot became *queryable*. The aligned value on bar i is $P^\star(t_i) = P(\tau_{j^\star})$ with $j^\star = \max\{j : \tau_j \leq t_i - \delta\}$, where $\delta \geq 0$ is a mandatory latency guard.

The join is strictly backward-looking and carries an explicit latency guard δ absorbing ingestion, analysis, and publication delay. Two failure modes are thereby excluded by construction: using a snapshot timestamped before it was computable (*look-ahead*), and using the revised value of a snapshot later corrected (*restatement*); the source must expose the value as first known, not as later amended.

4.3 Reusing the five-gate replication protocol

Once aligned, a narrative column traverses the same protocol as any candidate (Onat, 2026b): (1) a *discovery* gate testing information coefficient against forward returns across horizons under Benjamini–Hochberg control; (2) a *cost* gate re-testing net of realistic fees, funding, and slippage; (3) a *temporal* gate requiring persistence across disjoint time folds; (4) a *cross-symbol* gate requiring the effect to replicate across entities that map to distinct concepts; and (5) a *correlation* gate testing incremental content against the incumbent book. A narrative signal that fails any gate is discarded exactly as an endogenous one would be. Crucially, the horizons at which narrative is tested are longer than those for microstructure—hours to days, not seconds—because that is where Section 5 predicts its content lies; the gate thresholds and FDR ledger are shared, the horizon grid is not.

5 Why It Helps I: Timescale Separation

Proposition 5.1 (Near-orthogonality by band disjointness). Let $x(t)$ be an order-book-imbalance signal whose predictive cross-spectrum with returns is supported on a high-frequency band B_x (empirically $\approx [0.005, 0.1]$ Hz on this venue), and let $\dot{P}(t)$ be narrative momentum sampled at cadence Δ with $1/\Delta \ll \min B_x$, so that its spectral content is supported on a low-frequency band B_P with $B_P \cap B_x = \emptyset$. If both are approximately band-limited to their supports, their contemporaneous correlation obeys

$$|\text{corr}(x, \dot{P})| \leq \frac{\int_{B_x \cap B_P} \sqrt{S_x(f) S_{\dot{P}}(f)} df}{\sqrt{\int S_x \int S_{\dot{P}}}} \approx 0,$$

by the Cauchy–Schwarz bound on cross-spectral density over disjoint supports.

The content of Proposition 5.1 is not that narrative and microstructure are provably uncorrelated—real signals are only approximately band-limited, and spectral leakage leaves a small residual—but that their correlation is *structurally suppressed*: they carry information at different frequen-

cies, so a book combining them is combining near-independent bets rather than repackaging the same one. This is precisely the property that the effective-breadth argument rewards.

Remark 5.1 (Why the venue’s spectral result matters). The empirical localization of imbalance predictability to a narrow high-frequency band (companion spectral study) is what makes the disjointness concrete: it is not merely that narrative is “slower” in the abstract, but that the incumbent book’s information demonstrably vanishes above periods of a few hundred seconds, exactly where a narrative signal sampled every few minutes to hours begins.

6 Why It Helps II: Effective Breadth

We formalize the marginal value of a low-correlation signal. Consider a book of N microstructure signals, each standardized, with common individual information coefficient IC and common pairwise correlation $\rho \in [0, 1)$. Its effective breadth is

$$N_{\text{eff}}(N, \rho) = \frac{N}{1 + (N - 1)\rho}, \quad (1)$$

and by the fundamental law $IR \approx IC \sqrt{N_{\text{eff}}}$.

Proposition 6.1 (A low-correlation signal buys more breadth). Append one further signal of equal individual strength IC that has correlation ρ_x with each incumbent. Write the augmented effective breadth as $N'_{\text{eff}}(\rho_x)$ computed from the $(N+1) \times (N+1)$ correlation matrix with incumbent block correlation ρ and new-row correlation ρ_x . Then N'_{eff} is strictly decreasing in ρ_x ; consequently an exogenous signal with $\rho_x \approx 0$ (Proposition 5.1) delivers strictly greater marginal effective breadth than an additional microstructure signal with $\rho_x = \rho > 0$.

Sketch. The mean off-diagonal correlation of the augmented book is $\bar{\rho}' = [N(N-1)\rho + 2N\rho_x] / [(N+1)N]$, which is increasing in ρ_x ; the average-correlation form of the effective breadth, $N'_{\text{eff}} = (N+1) / [1 + N\bar{\rho}']$, is decreasing in $\bar{\rho}'$ and hence in ρ_x . The final claim is the comparison at $\rho_x \approx 0$ versus $\rho_x = \rho$. $\square$

Proposition 6.1 is the quantitative statement of “more complete.” Beyond some depth, the $N+1$ -th microstructure signal is nearly redundant—it raises N but also $\bar{\rho}$, and N_{eff} barely moves—whereas the first signal from a new, near-orthogonal domain moves N_{eff} by close to a full unit even at modest individual IC . A book’s information ratio is grown far more cheaply by *widening its domains* than by *deepening one domain*.

Remark 6.1 (Stress-regime correlation is the binding quantity). The correlations in (1) must be measured where diversification is actually needed: in stress. Signals that are decorrelated in calm markets frequently co-move when volatility spikes, collapsing realized breadth exactly when it is demanded. A narrative signal is a candidate to break this pattern precisely because its driver—public discourse—is not the order flow that convulses in a microstructure stress event; but this is a hypothesis, and the evaluation therefore estimates ρ_x *conditionally on volatility regime*, not marginally.

7 Why It Helps III: The Adverse-Selection Barrier

The sharpest limit on microstructure alpha is not weak prediction but adverse selection: the signal predicts the future, yet a directionally-correct passive order is filled preferentially in the states where the prediction is wrong. Formally, for signal g and forward return r , let F be the event that a passive order placed on g ’s side is filled. The exploitable skill is the *conditional* information coefficient $IC(g, r \mid F)$, and the barrier is the gap $IC(g, r) - IC(g, r \mid F)$, which the companion study finds to be almost the whole of IC for order-book imbalance.

Proposition 7.1 (Slow signals couple weakly to the fill state). Suppose signal g is exploited not by a reactive passive quote at the touch but by a schedule that works into the position over a horizon H_g comparable to g ’s own decay timescale. Then the relevant conditioning event is not the instantaneous touch-fill F but the average execution outcome over $[t, t + H_g]$, whose dependence on the fast fill-selection mechanism is diluted by a factor growing with H_g relative to the spread-crossing timescale τ_s . Hence the conditional-IC degradation of a signal with $H_g \gg \tau_s$ is bounded above by that of a signal with $H_g \approx \tau_s$, and can be strictly smaller.

The intuition is executional. Microstructure edge at second horizons must be captured by winning passive fills in a race, and the winner of that race is systematically the adversely-selected side. A narrative signal with an hours-to-days horizon is captured differently: one has time to work orders patiently, cross spreads selectively, and split execution, so the pathological correlation between “getting filled” and “being wrong” is materially weaker. Proposition 7.1 does not claim the barrier vanishes—exogenous signals face their own frictions, including that a widely-read narrative may already be priced—but it identifies a concrete channel by which a slow signal can convert a larger fraction of its statistical content into economic content than a fast one.

8 Evaluation Protocol

Each claim above is falsifiable; we specify the test and its kill criterion, and report no numbers.

1. **Orthogonality (tests Prop. 5.1).** Estimate the volatility-regime-conditional correlation ρ_x between aligned narrative momentum and the incumbent imbalance signal, and the cross-spectral coherence across frequency. *Kill:* if ρ_x is not small in stress regimes, the effective-breadth argument does not apply and the signal offers no diversification.
2. **Incremental predictive content (tests the reason to include at all).** Estimate the conditional mutual information $I(r_{t \rightarrow t+h}; \dot{P}_t \mid \text{price}_t, \text{volume}_t)$ at narrative-scale horizons h , under the k NN estimator with a block permutation null and Benjamini–Hochberg control across concepts and horizons. *Kill:* if conditional MI is not significantly positive after FDR, narrative adds nothing beyond price and volume.
3. **Breadth realization (tests Prop. 6.1).** Compute the book’s N_{eff} and out-of-sample information ratio with and without the narrative signal, walk-forward. *Kill:* if N_{eff} and OOS IR do not rise on inclusion, the marginal-breadth claim fails in practice.
4. **Execution realizability (tests Prop. 7.1).** Under a schedule-based execution model at horizon H_g , estimate the conditional information coefficient $\text{IC}(\dot{P}, r \mid F)$ and compare its degradation to that of the imbalance signal. *Kill:* if the conditional IC collapses as severely as for microstructure, the adverse-selection advantage is illusory.
5. **Leakage and stability audit.** Re-run studies 1–4 with the latency guard δ inflated and with the entity map ϕ frozen strictly before the test window. *Kill:* if results depend on a short δ or on post-hoc concept selection, they are artifacts.

Study 2 is the necessary condition—without incremental information there is nothing to integrate—and studies 3 and 4 are what distinguish a signal that merely predicts from one that improves a *book*. All five are framed as leakage-controlled and multiple-testing-corrected because a signal tested against many concepts and horizons is otherwise a textbook data-snooping trap.

9 What “More Complete” Means

The object this paper argues for is not a narrative strategy but a *multi-domain, decorrelated alpha book that reports its own honest effective breadth*. In that book the microstructure signals remain the high-frequency core; the narrative signal is a low-frequency, near-orthogonal overlay admitted through the same gates and deployed as one more process, not as a parallel system. Three consequences follow. The book’s information ratio can grow through width rather than through ever more correlated depth (Section 6). Its failure modes diversify: a microstructure regime break need not coincide with a narrative regime break (Remark 6.1). And a portion of its statistical content may survive the executional filter that caps its order-flow signals (Section 7). Sophistication here is not additional complexity for its own sake but the disciplined admission of a second information domain under an unchanged standard of proof—the platform becomes more complete by knowing, and gating, what it does not otherwise see.

10 Limitations

We disclose the principal threats. The narrative signal inherits every measurement limitation of its instrument (Onat, 2026a): language-model sentiment error, lexical rather than semantic clustering, and frequently-inert engagement terms; these bound the fidelity of $\dot{P}$ and hence of any measured effect. Narrative events are *sparse* relative to microstructure ticks, so statistical power at narrative horizons is limited and the multiple-testing correction is correspondingly punishing—an honest evaluation may simply fail to reject the null for want of events. The narrative–return relationship is plausibly *non-stationary*: a signal that leads price while under-attended can decay to contemporaneous as it becomes widely read, so any positive finding requires ongoing decay monitoring rather than a one-time validation. The entity map ϕ is a researcher choice and a degree of freedom that must be frozen before testing. Finally, the theoretical propositions are deliberately modest: Proposition 5.1 bounds correlation only for approximately band-limited signals, Proposition 6.1 assumes equal individual skill for its clean comparison, and Proposition 7.1 identifies a channel without quantifying it. None of these is a claim of profit; each is a signpost for the evaluation.

11 Conclusion

We have set out how, and why, to admit an exogenous narrative signal into an autonomous microstructure alpha book. The how is an integration architecture that renders narrative indistinguishable from an endogenous feature to the scoring machinery, aligned point-in-time and gated by an unchanged five-gate protocol. The why is three structural arguments—timescale separation making the signal near-orthogonal by construction, effective breadth rewarding that orthogonality more than additional correlated depth, and a weaker coupling to the fill state partially evading the adverse-selection barrier that caps order-flow alpha. Each is a hypothesis with a stated, leakage-controlled, FDR-corrected test and an explicit kill criterion; none is a performance claim. The contribution is a principled route to a more complete quantitative research platform: not a bigger pile of order-book features, but a second information domain, admitted under the same standard of proof, whose value—if it exists—shows up exactly as increased effective breadth and surviving conditional skill. Turning the protocol on data is the next step.

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